

PRACTICE TEST

**Science and
Technology/Engineering
Grade 5**

Student Name

School Name

District Name

Grade 5 Science and Technology/Engineering PRACTICE TEST

This practice test contains 23 questions.

Directions

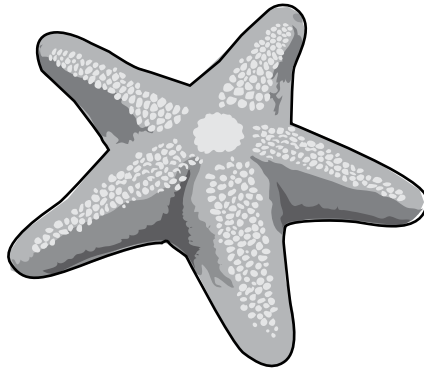
Read each question carefully and then answer it as well as you can. You must record all answers in this Practice Test Booklet.

For some questions, you will mark your answers by filling in the circles in your Practice Test Booklet. Make sure you darken the circles completely. Do not make any marks outside of the circles. If you need to change an answer, be sure to erase your first answer completely.

If a question asks you to show or explain your work, you must do so to receive full credit. Write your response in the space provided in this Practice Test Booklet. Only responses written within the provided space will be scored.

If you do not know the answer to a question, you may go on to the next question. When you are finished, you may review your answers and go back to any questions you did not answer.

- 1 A group of students observe some sea stars (also known as starfish) while on a field trip. A sea star is shown in the picture below.



Most of the sea stars have five arms, but two of the sea stars each have an arm that is much shorter than their other arms. A park ranger tells the students that the shorter arms were once missing because they were probably eaten by a fish or a crab, but the arms have started to grow back.

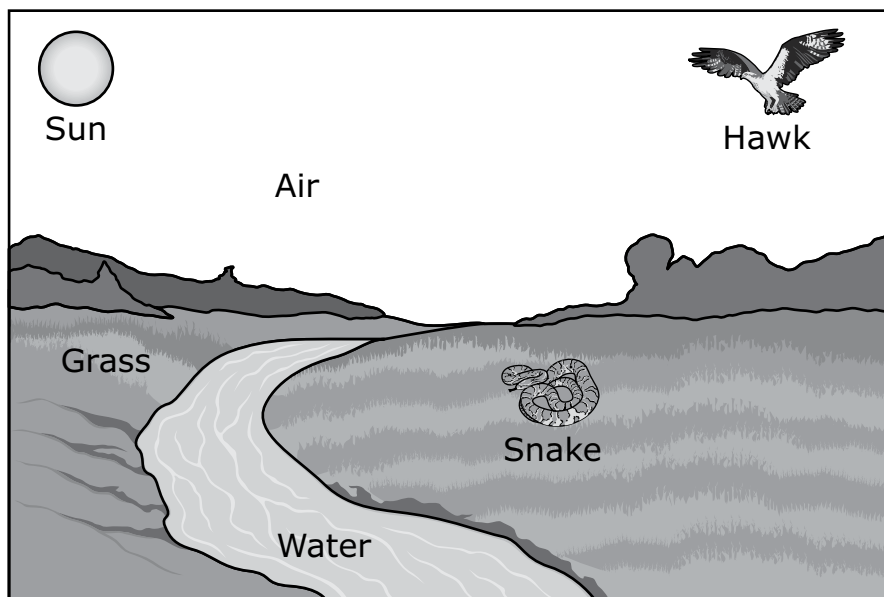
Which of the following **best** describes how sea stars are able to regrow their arms?

- Ⓐ Sea stars learn how to regrow their arms.
- Ⓑ Sea stars inherit the ability to regrow their arms.
- Ⓒ Sea stars eat food that causes their arms to regrow.
- Ⓓ Sea stars move to warmer water that causes their arms to regrow.

- 2** A student took pictures of the Moon on different days. Which set of pictures did the student take on Sunday, Wednesday, and Saturday of the same week?



- 3 The diagram shows parts of an ecosystem.

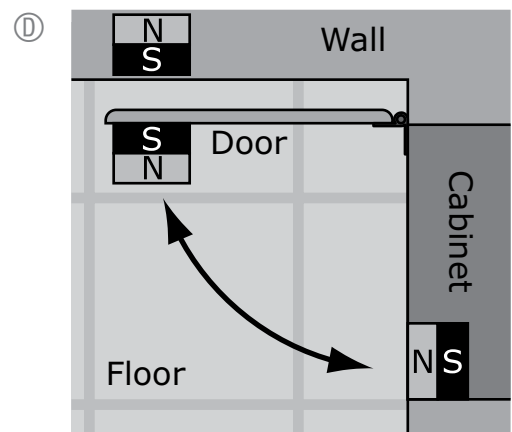
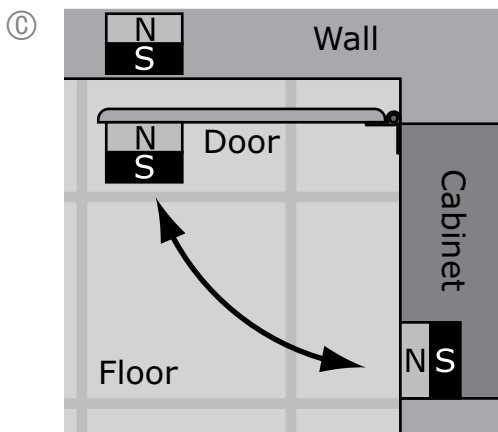
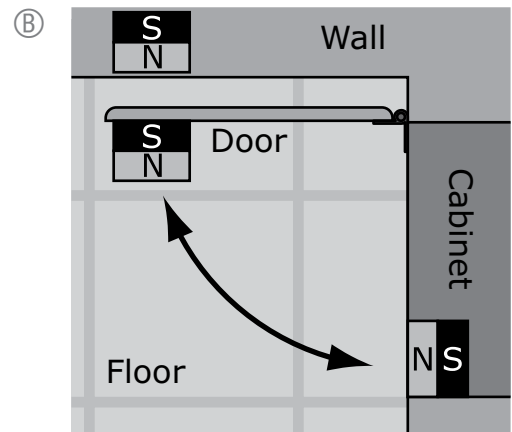
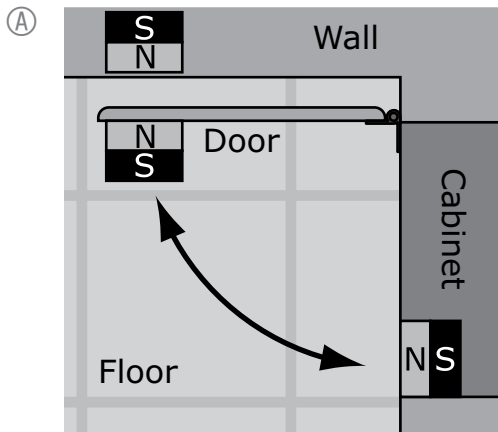


Which of the following is the source of all the energy in the ecosystem?

- Ⓐ Air
- Ⓑ Grass
- Ⓒ Hawk
- Ⓓ Snake
- Ⓔ Sun
- Ⓕ Water

- 4** A carpenter wants to use magnets to help keep a cabinet door closed. The carpenter also wants to make sure the cabinet door does not hit a nearby wall when the cabinet door is opened.

Which diagram shows how the carpenter should place the magnets?



- 5** A student is testing different types of wood to use for a bookshelf. The student tests each type of wood to predict how many books the bookshelf will safely hold.

Which of the following characteristics of the wood determines how many books the bookshelf can safely hold?

- Ⓐ flexibility
- Ⓑ hardness
- Ⓒ strength
- Ⓓ weight

The following section focuses on seasonal climate data.

Read the information below and use it to answer the three selected-response questions and one constructed-response question that follow.

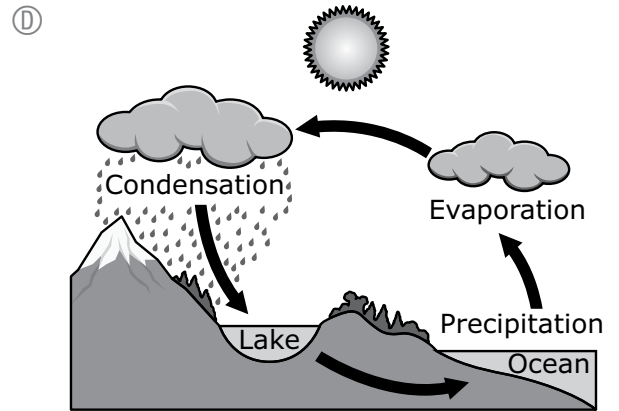
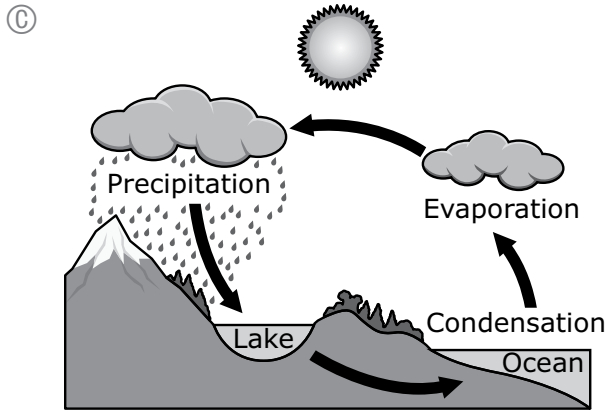
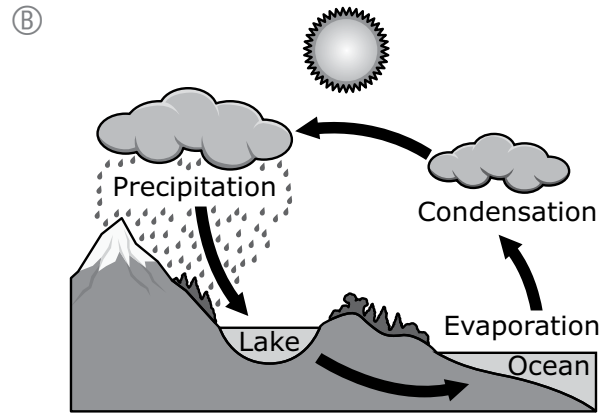
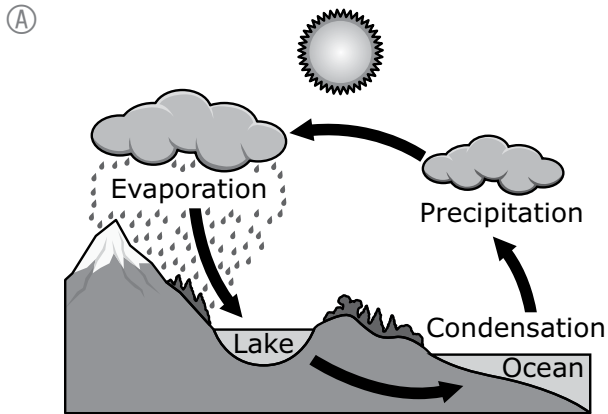
On a spring day, a student in Worcester, Massachusetts, reads in the news that a local river has flooded. The student asks a science teacher what caused the river to flood. The teacher explains that the flooding is a result of seasonal weather and the water cycle. The water cycle describes how water moves from one location to another. For example, water in an ocean may move through the water cycle and become part of a lake. As water moves and changes form, it has an effect on both daily and seasonal weather in an area.

The table shows seasonal climate data for Worcester, Massachusetts.

Seasonal Climate Data for Worcester, MA

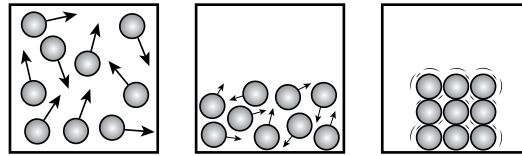
Season	Average Precipitation (inches)	Average Low Temperature (°F)	Average High Temperature (°F)	Average Wind Speed (miles per hour)
winter	10.5	19.6	34.0	11.4
spring	12.5	36.7	54.6	11.0
summer	12.1	59.3	76.8	8.6
fall	12.9	42.5	58.5	9.3

- 6** Which of the following diagrams correctly shows how water can move through the water cycle from an ocean to a lake?

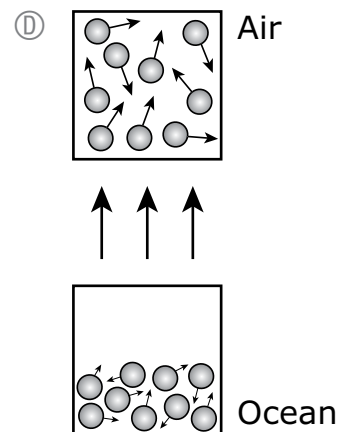
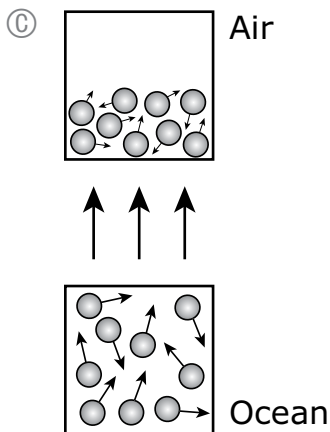
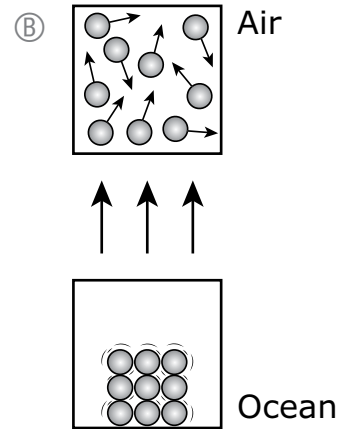
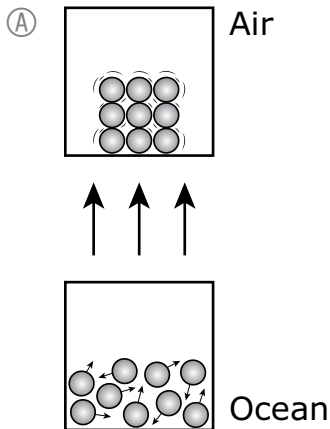


- 7 Which of the following **best** explains why more water flows in a river in Worcester in spring than in other seasons?
- Ⓐ The melting snow in spring causes more runoff.
 - Ⓑ The high winds in spring push on the river at a greater speed.
 - Ⓒ The average high temperature in spring results in fewer rainstorms.
 - Ⓓ The average low temperature in spring causes less water to evaporate from the river.

- 8 The diagram shows water particles in three different phases.



On a summer day, water particles from the ocean change phase as they move into the air. Which of the following models best shows this process?

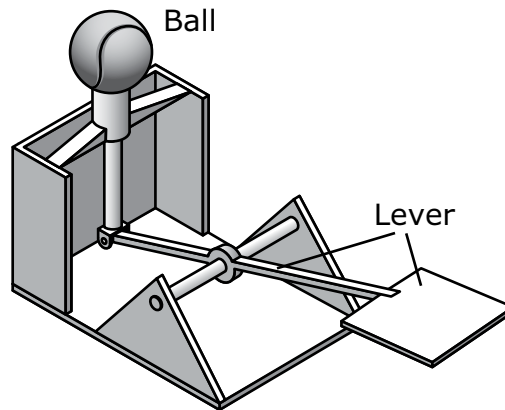


This question has two parts. Write your response on the next page. Be sure to label each part of your response.

- 9** The amounts of water on and below Earth's surface can change during the year.
- A. Identify the season when the **least** amount of water becomes groundwater in Worcester. Explain why this season has the least amount of groundwater. Include data from the seasonal climate table to support your answer.
 - B. Identify the season when the **greatest** amount of water moves from Earth's surface into the atmosphere in Worcester. Explain why this season has the greatest amount of water moving into the atmosphere. Include data from the seasonal climate table **and** describe the role of energy to support your answer.

9

- 10 A dog trainer designed a device to launch a ball into the air for a dog to catch. The device launches the ball when a sandbag is dropped on the lever shown.



The trainer wants to be able to launch the ball one meter into the air.

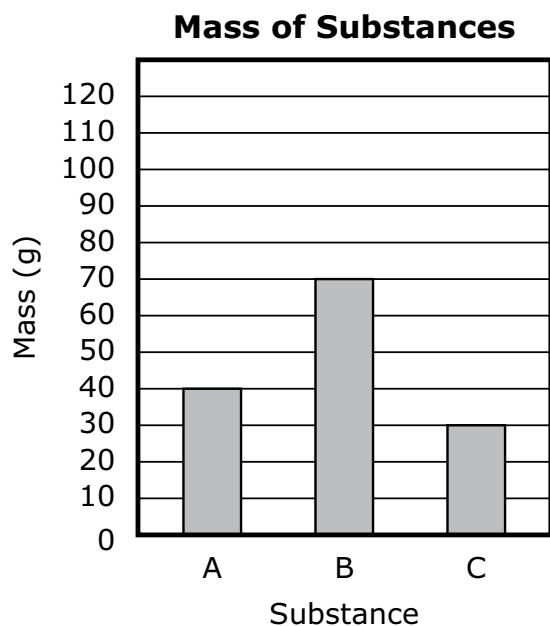
Which set of steps should the dog trainer follow to make sure the device launches the ball to the correct height?

- Ⓐ
 1. Drop the sandbag.
 2. Listen for a sound.
 3. Replace the sandbag with something that weighs the same.
 4. Repeat all steps until the ball reaches exactly one meter.
- Ⓑ
 1. Drop the sandbag.
 2. Measure how high the ball reaches.
 3. Replace the sandbag with something that has a different shape.
 4. Repeat all steps until the ball reaches exactly one meter.
- Ⓒ
 1. Drop the sandbag.
 2. Add electrical energy to the system.
 3. Replace the sandbag with something that weighs the same.
 4. Repeat all steps until the ball reaches exactly one meter.
- Ⓓ
 1. Drop the sandbag.
 2. Measure how high the ball reaches.
 3. Adjust the energy of the sandbag by holding it at a different height.
 4. Repeat all steps until the ball reaches exactly one meter.

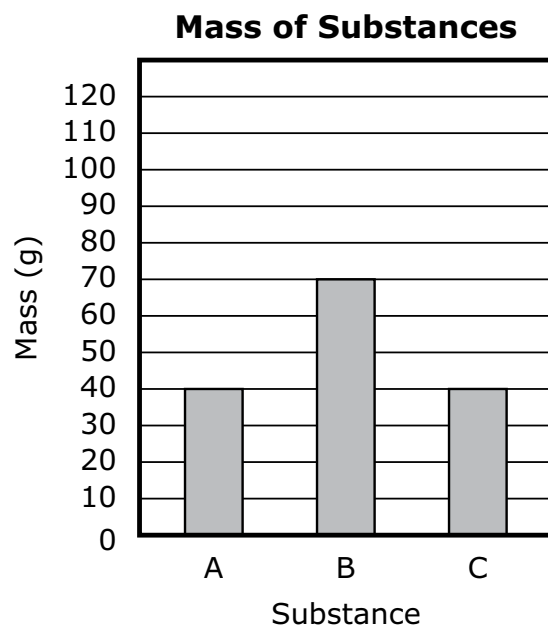
- 11** Two substances, substance A and substance B, react to form substance C. Substance A has a mass of 40 g, and substance B has a mass of 70 g. The reaction takes place in a closed system.

Which of the following graphs best represents the mass of each substance?

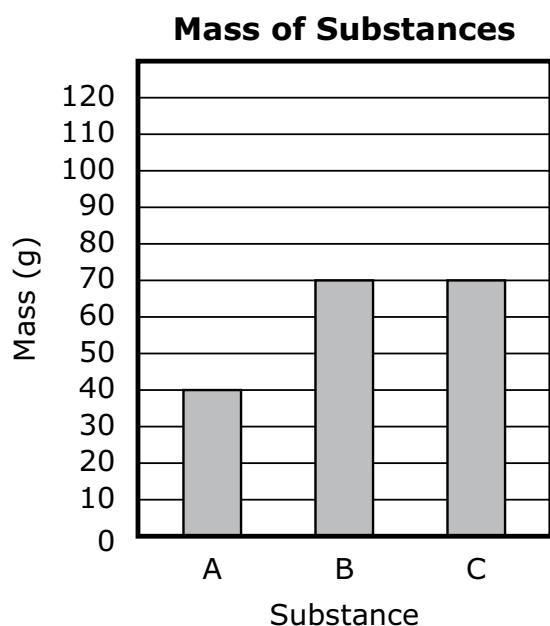
(A)



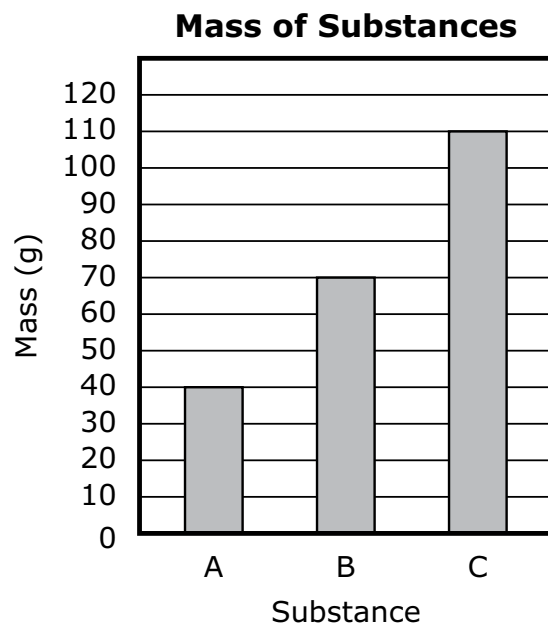
(B)



(C)



(D)

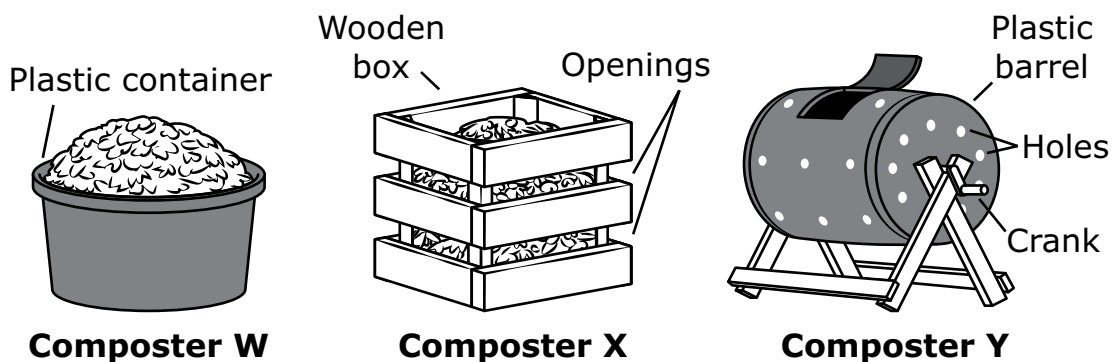


This question has three parts. Write your response on the next page. Be sure to label each part of your response.

12 Students are comparing three composters.

- Composter W is a wide, black plastic container with an open top.
- Composter X is a light-colored wooden box that has openings on all sides and an open top.
- Composter Y is a black plastic barrel with a door. The barrel has holes and can spin when the crank is turned.

The composters are shown.

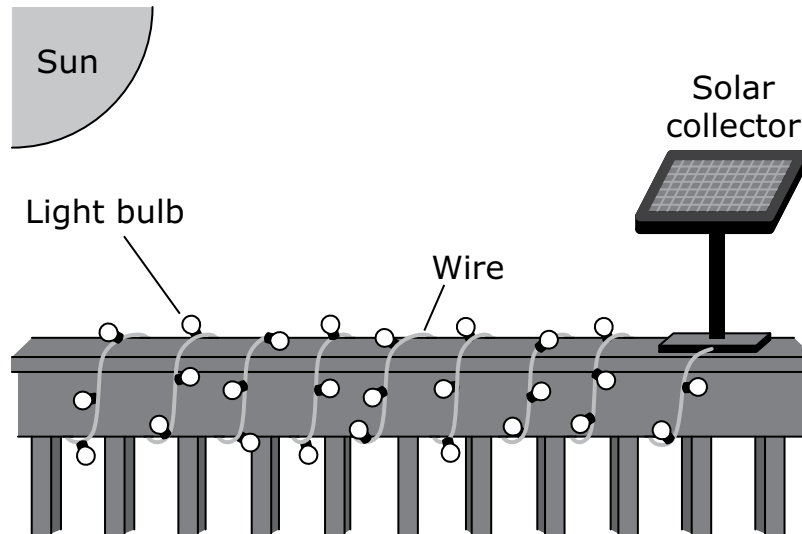


- A. Describe the purpose of a composter.
- B. Identify **two** conditions that make a composter work well.
- C. Identify which composter (W, X, or Y) will work best. Explain your answer using the conditions you identified in Part B.

12

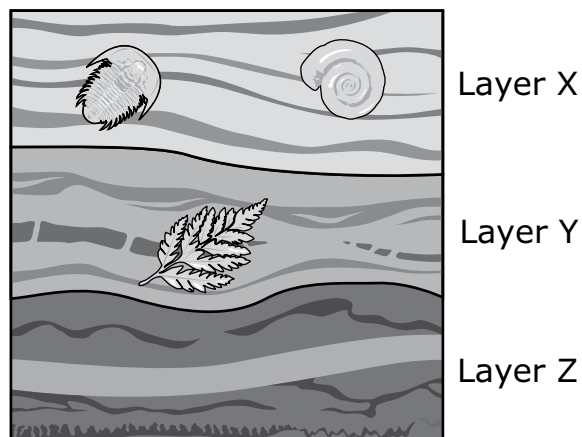
Write your response on the next page. Be sure to label each part of your response.

- 13** Solar string lights are sometimes used as a decoration. An example of solar string lights is shown.



The solar collector is connected by a wire to the light bulbs. Describe **two** ways in which energy is transferred in the setup. In your description, include the types of energy that are transferred.

- 14 A cross section of three sedimentary rock layers is shown.



Which of the following represents the rock layers in order from the youngest to the oldest?

- Ⓐ Layer X → Layer Y → Layer Z
- Ⓑ Layer X → Layer Z → Layer Y
- Ⓒ Layer Y → Layer Z → Layer X
- Ⓓ Layer Z → Layer Y → Layer X

This question has two parts.

- 15** The characteristics of trees can be influenced by the environment and by inheritance.

Part A

Determine whether each characteristic is the result of the environment, inheritance, or both the environment and inheritance.

The height of a tree in a forest is the result of

- Ⓐ the environment.
- Ⓑ inheritance.
- Ⓒ both the environment and inheritance.

A burn mark on a tree in a forest is the result of

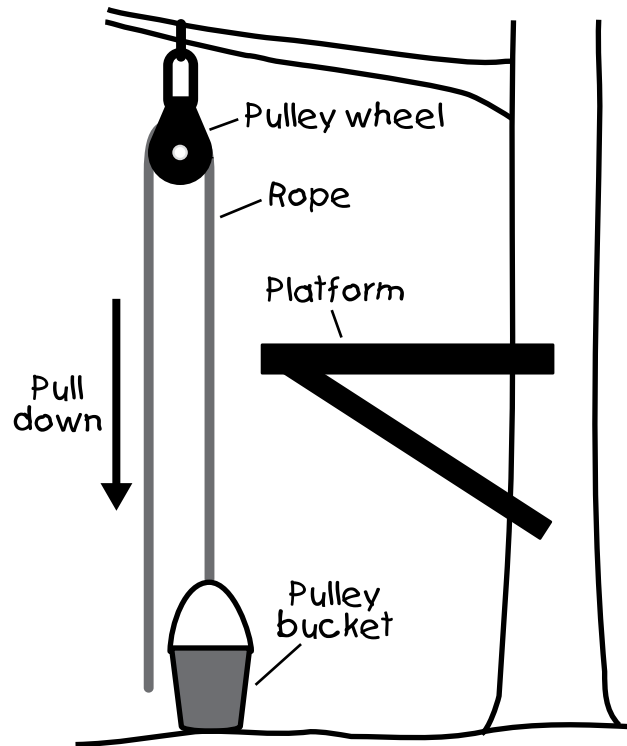
- Ⓐ the environment.
- Ⓑ inheritance.
- Ⓒ both the environment and inheritance.

Part B

Inherited characteristics of trees come from which of the following?

- Ⓐ air particles
- Ⓑ parent trees
- Ⓒ nutrients from soil
- Ⓓ several nearby trees

- 16 The diagram shows a student's design for a pulley system used to lift objects up to a platform.



Which of the following is **most** important to consider when choosing the type of rope for the pulley system?

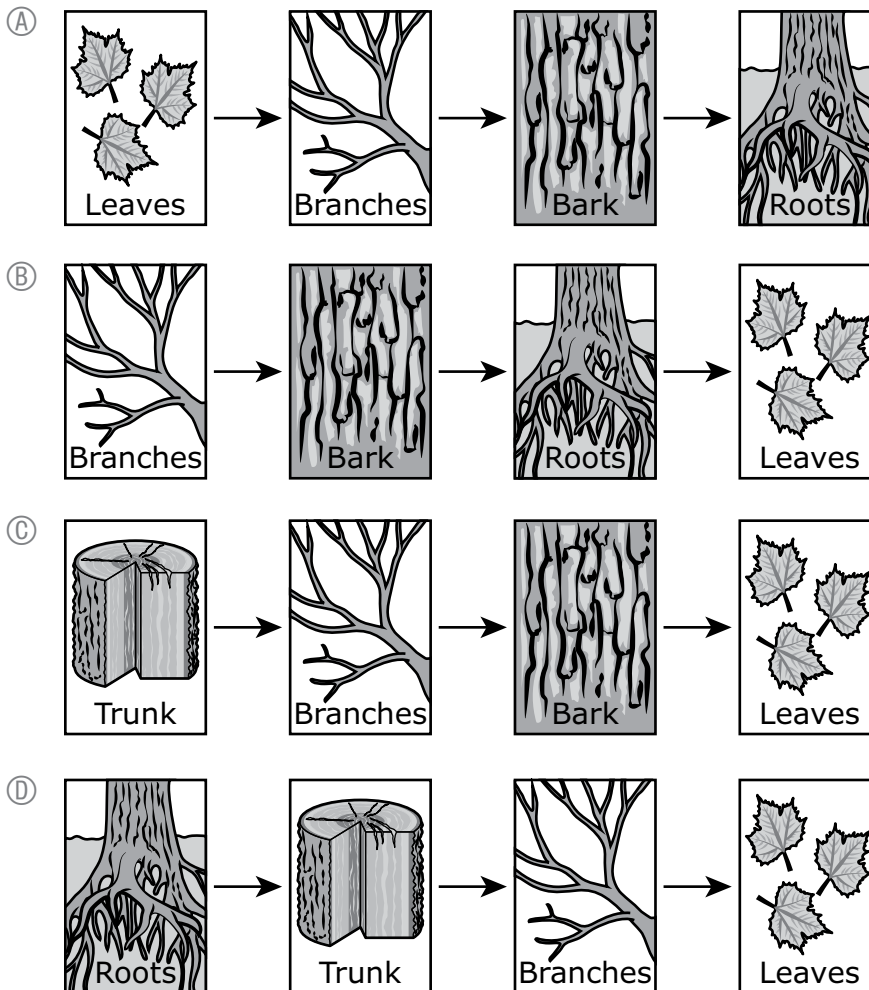
- Ⓐ the shape of the pulley bucket handle
- Ⓑ the speed at which the pulley wheel moves
- Ⓒ the amount of weight the pulley bucket will hold
- Ⓓ the distance of the pulley wheel from the platform

This question has two parts.

17 Part A

After water enters a tree, it travels through different parts of the tree.

Which of the following best shows the order in which water moves through the tree's parts?



Part B

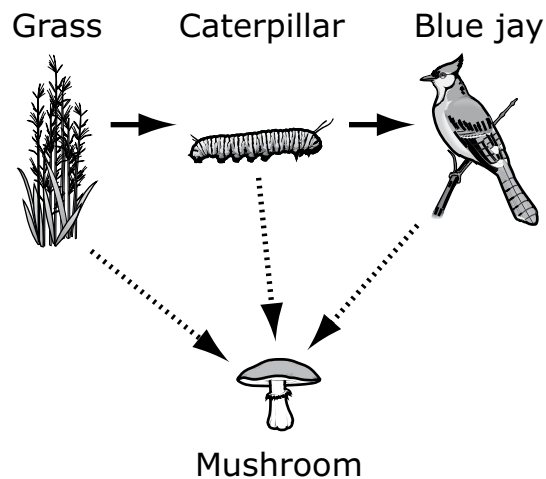
What is the main reason the tree needs structures to move water?

- (A) to protect the tree from insects
- (B) to support the tree in strong winds
- (C) to make sugars so the tree can grow

18 Select the **two** sentences that describe examples of erosion.

- Ⓐ Ice melts on a lake.
- Ⓑ Waves rise and fall in the ocean.
- Ⓒ Rainwater moves soil down a hill.
- Ⓓ Rock forms at the bottom of the ocean.
- Ⓔ Wind blows sand on a beach to a different area.

- 19 A food web is shown.



Which of the following best classifies each organism based on its role in the food web?

Ⓐ

Producer	Consumer	Decomposer
grass mushroom	blue jay	caterpillar

Ⓑ

Producer	Consumer	Decomposer
grass	blue jay caterpillar	mushroom

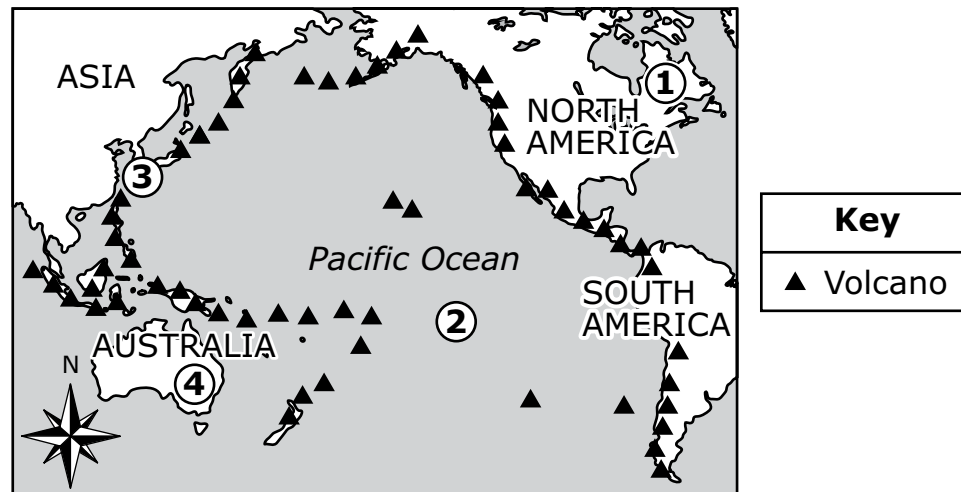
Ⓒ

Producer	Consumer	Decomposer
blue jay caterpillar	grass	mushroom

Ⓓ

Producer	Consumer	Decomposer
blue jay	mushroom	grass caterpillar

- 20 Four locations on the map are labeled 1, 2, 3, and 4.



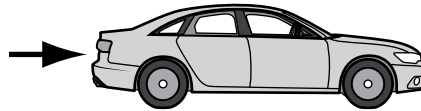
Which location is most likely on a plate boundary?

- 21 A student is constructing four different electrical circuits. Each circuit has a battery, wires, a switch, and a light bulb.

Which of the following is the **best** way for the student to record information so that another student can construct exactly the same four circuits?

- (A) Draw a diagram that shows each circuit.
- (B) Make a tally of the parts used in each circuit.
- (C) Make a list of the materials needed for each circuit.
- (D) Create a graph that shows the number of parts in each circuit.

- 22** The model shows a force acting on a car. The amount of the force is represented by the size of the arrow.



A second force is applied to the car. As a result, the car's motion stops changing.

Which of the following models best represents the forces acting on the car as the second force is applied?

- (A) A side-view illustration of a car. Two arrows of equal length point horizontally in opposite directions: one to the rear and one to the front.
- (B) A side-view illustration of a car. Two arrows of equal length point horizontally in opposite directions: one to the rear and one to the front.
- (C) A side-view illustration of a car. Two arrows point horizontally in opposite directions: one to the rear and one to the front. The arrow pointing to the front is significantly longer than the arrow pointing to the rear.
- (D) A side-view illustration of a car. Two arrows point horizontally in opposite directions: one to the rear and one to the front. The arrow pointing to the front is significantly longer than the arrow pointing to the rear.

- 23 Butterflies go through different stages of life, including birth, growth, and reproduction. Which of the following best classifies four events that occur during those stages?

(A)

Birth	Growth	Reproduction
Adults mate and the females lay eggs.	Larvae change form and become adults. Larvae get older and bigger.	Larvae hatch from eggs.

(B)

Birth	Growth	Reproduction
Larvae get older and bigger.	Larvae change form and become adults.	Adults mate and the females lay eggs. Larvae hatch from eggs.

(C)

Birth	Growth	Reproduction
Larvae hatch from eggs.	Larvae change form and become adults. Larvae get older and bigger.	Adults mate and the females lay eggs.

(D)

Birth	Growth	Reproduction
Larvae change form and become adults. Adults mate and the females lay eggs.	Larvae get older and bigger.	Larvae hatch from eggs.

MCAS Grade 5 Science and Technology/Engineering (STE) Paper-Based Practice Test Answer Key

The practice test is approximately equal to the number of questions students experience in a single session of the MCAS Grade 5 STE test. Information about the test design posted on the DESE [website](#). The following pages include the reporting category, alignment to standards from the [Curriculum Frameworks](#), and practice (if applicable) for each question on the practice test. Answers are provided for selected-response items, and rubrics and sample student responses are included for constructed-response items.

PBT Item No.	Reporting Category	Standard	Science and Engineering Practice Category	Item Type*	Max Points	Correct Answer**
1	<i>Life Science</i>	3.LS.3.2	No Practice	SR	1	B
2	<i>Earth and Space Science</i>	5.ESS.1.2	C. Evidence, Reasoning, and Modeling	SR	1	C
3	<i>Life Science</i>	5.PS.3.1	C. Evidence, Reasoning, and Modeling	SR	1	E
4	<i>Physical Science</i>	3.PS.2.4	C. Evidence, Reasoning, and Modeling	SR	1	A
5	<i>Physical Science</i>	5.PS.1.3	No Practice	SR	1	C
6	<i>Earth and Space Science</i>	5.ESS.2.1	C. Evidence, Reasoning, and Modeling	SR	1	B
7	<i>Earth and Space Science</i>	5.ESS.2.1	No Practice	SR	1	A
8	<i>Physical Science</i>	5.PS.1.1	C. Evidence, Reasoning, and Modeling	SR	1	D
9	<i>Earth and Space Science</i>	3.ESS.2.2	B. Mathematics and Data	CR	3	<i>see page 3</i>
10	<i>Technology/Engineering</i>	4.ETS.1.3	A. Investigations and Questioning	SR	1	D
11	<i>Physical Science</i>	5.PS.1.2	B. Mathematics and Data	SR	1	D
12	<i>Life Science</i>	5.LS.2.2	C. Evidence, Reasoning, and Modeling	CR	3	<i>see page 4</i>
13	<i>Physical Science</i>	4.PS.3.2	C. Evidence, Reasoning, and Modeling	CR	2	<i>see page 5</i>
14	<i>Earth and Space Science</i>	4.ESS.1.1	C. Evidence, Reasoning, and Modeling	SR	1	A
15	<i>Life Science</i>	3.LS.3.2	No Practice	SR	2	Part A: C,A Part B: B
16	<i>Technology/Engineering</i>	4.ETS.1.5	A. Investigations and Questioning	SR	1	C
17	<i>Life Science</i>	4.LS.1.1	C. Evidence, Reasoning, and Modeling	SR	2	Part A: D Part B: C
18	<i>Earth and Space Science</i>	4.ESS.2.1	No Practice	SR	1	C,E
19	<i>Life Science</i>	5.LS.2.1	C. Evidence, Reasoning, and Modeling	SR	1	B
20	<i>Earth and Space Science</i>	4.ESS.2.2	C. Evidence, Reasoning, and Modeling	SR	1	C

21	<i>Technology/ Engineering</i>	3.ETS.1.4	No Practice	SR	1	A
22	<i>Physical Science</i>	3.PS.2.1	C. Evidence, Reasoning, and Modeling	SR	1	B
23	<i>Life Science</i>	3.LS.1.1	C. Evidence, Reasoning, and Modeling	SR	1	C

* Science item types are selected-response (SR) and constructed-response (CR).

**Pages 3–5 of this document provide sample student responses to constructed-response items.

Sample Student Responses for Constructed-Response Items

Item 9 Sample Student Work and Scoring Guide

Scoring Guide

Score	Description
3	The response demonstrates a thorough understanding of using the cycling of water through a watershed. The response correctly identifies the season when the least amount of water becomes groundwater, clearly explains why this season has the least amount of groundwater, and uses data from the table to support why it is that season. The response correctly identifies the season when the greatest amount of water moves from Earth's surface into the atmosphere, clearly explains why this season has the greatest amount of water moving into the atmosphere, and uses data from the table to support why it is that season.
2	The response demonstrates a partial understanding of using the cycling of water through a watershed.
1	The response demonstrates a minimal understanding of using the cycling of water through a watershed.
0	The response is incorrect or contains some correct work that is irrelevant to the skill or concept being measured.

Score Point 3

Part	Student Response
A	Winter has the least amount to groundwater beacause it doesn't have a lot of precipitation in the winter so the groundwater is very little beacuse only 10.5 inches of precipitation fall usually every year in the winter.
B	Summer is beacuse it has just 0.4 more inches of rain and it has the hottest tempatures so evaporation is going to happen a lot. Heat energy is going to make the 12.1 inches of precipitation evaporate.

Score Point 2

Part	Student Response
A	Winter has the least amount of water that becomed ground water because the temperatures get so low tat the water will just freeze into a solid and you would have to burry all the ice to make it go into the ground. <i>(Note: Student did not provide data to support answer.)</i>
B	Summer has the greatest amount of water that moves into the atmosphere because the temperatures will get high enough to make the water evaporate and go into the atmosphere as part of the water cycle.

Score Point 1

Part	Student Response
A	A season that has the least amount of water is winter because in the winter its very cold and it snows a lot. So because it is so cold the snow doesn't melt at all. And also there wont be as much water in the winter as the other seasons.
B	The season with the greatest amount of water is fall beacuse in fall it rains a ton! Fall is just the begening of the cold weather so instead of snowing it rains.

Score Point 0

Part	Student Response
A	Summer because it is humid hot and dry but in winter it is snowy and a lot of ice and cold but in the spring it all ways rainy but in fall it is just rainy and cold so summer is it.
B	Spring because it always is rainy and it is very wet.

Item 12 Sample Student Work and Scoring Guide

Scoring Guide

Score	Description
3	The response demonstrates a thorough understanding of composter designs and their functions. The response clearly describes the purpose of a composter and correctly identifies two conditions that make a composter work well. The response also correctly identifies a composter design that will work best and clearly explains the answer.
2	The response demonstrates a partial understanding of the function of composter designs and their functions.
1	The response demonstrates a minimal understanding of the function of composter designs and their functions.
0	The response is incorrect or contains some correct work that is irrelevant to the skill or concept being measured.

Score Point 3

Part	Student Response
A	The purpose of a composter is to recycle food scraps and some other things that you would normally throwaway, sending them back into the Earth or a good fertilizer producer if you have a garden.
B	Two conditions that make a composter work are, lot's of oxygen and also lot's of decomposers like worms and other bugs.
C	I think that Composter X will work the best. It has giant spaces for small things to get in and it provides lost's of oxygen. I don't think the others will work as well because Composter W won't get enough oxygen in and out and Composer Y doesn't get enogh oxygen in and out as well, but it has the holes for some small things to get in.

Score Point 2

Part	Student Response
A	A composter makes compost. Compost is a fertilizer that helps plants grow.
B	All composters need air, that is why they have holes or openings. They also need worms to transform the compost.
C	I think composter Y would work the best. It has air holes, a door to place worms, and a door to place other ingredients.

Score Point 1

Part	Student Response
A	To make compost which can be used to fertilize the soil.
B	spinning and turning
C	y, it can spin

Score Point 0

Part	Student Response
A	I think the purpose of a composter is to hold stuff, for example food, hay, animals, cement, or anything that can fit inside of it.
B	One condition is how it can hold a lot, another is its plastic and its not that easy to break.
C	I think the wide plastic container is the best because you can carry a lot of stuff, and its easy to carry around when your traveling to places.

Item 13 Sample Student Work and Scoring Guide

Scoring Guide

Score	Description
2	The response demonstrates a thorough understanding of energy being transferred from place to place by light, heat, and electric currents. The response clearly describes two ways in which energy is transferred in the setup and includes the types of energy that are transferred.
1	The response demonstrates a partial understanding of energy being transferred from place to place by light, heat, and electric currents.
0	The response is incorrect or contains some correct work that is irrelevant to the skill or concept being measured.

Score Point 2 Student Response

One way energy is transferred in this setup is that the Solar energy from the sun is collected by the solar panel and then turned into electrical energy. Another way that the energy is transferred is that the electrical energy in the wires turns into heat and light energy in the lightbulbs.

Score Point 1 Student Response

The energy from the sun is collected by the solar collector, and the energy the solar collector transfers energy to the bulbs. The light and heat energy from the sun get turned into electrical energy.

Score Point 0 Student Response

The solar collector transfers thermal energy from the sun to the wires, which will make light energy which will turn on the light bulbs.